

Data Collection and Preprocessing Phase

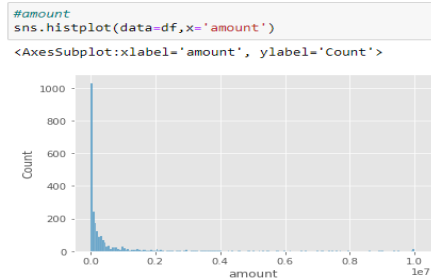
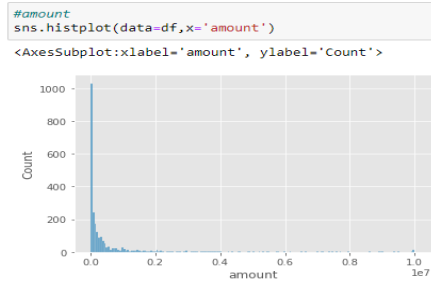
Date	15 JUNE 2025
Team ID	SWTID1749707061
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	6 Marks

Preprocessing Template

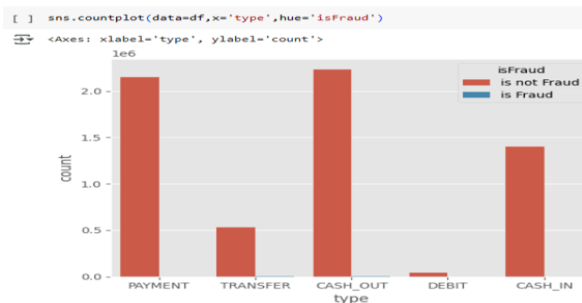
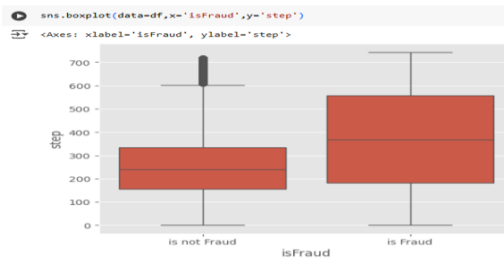
Dataset variables will be statistically analyzed to identify patterns and outliers, with Python employed for preprocessing tasks like normalization and feature engineering. Data cleaning will address missing values and outliers, ensuring quality for subsequent analysis and modeling, and forming a strong foundation for insights and predictions.

Section	Description				
Data Overview		step	amount	oldbalanceOrg	newbalanceOrig
	count	6.362620e+06	6.362620e+06	6.362620e+06	6.362620e+06
	mean	2.433972e+02	1.798619e+05	8.338831e+05	8.551137e+05
	std	1.423320e+02	6.038582e+05	2.888243e+06	2.924049e+06
	min	1.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
	25%	1.560000e+02	1.338957e+04	0.000000e+00	0.000000e+00
	50%	2.390000e+02	7.487194e+04	1.420800e+04	0.000000e+00
	75%	3.350000e+02	2.087215e+05	1.073152e+05	1.442584e+05
	max	7.430000e+02	9.244552e+07	5.958504e+07	4.958504e+07
		oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
	count	6.362620e+06	6.362620e+06	6.362620e+06	6.362620e+06
	mean	1.100702e+06	1.224996e+06	1.290820e-03	2.514687e-06
	std	3.399180e+06	3.674129e+06	3.590480e-02	1.585775e-03
	min	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
	25%	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
	50%	1.327057e+05	2.146614e+05	0.000000e+00	0.000000e+00
	75%	9.430367e+05	1.111909e+06	0.000000e+00	0.000000e+00
	max	3.560159e+08	3.561793e+08	1.000000e+00	1.000000e+00

Univariate Analysis



Bivariate Analysis

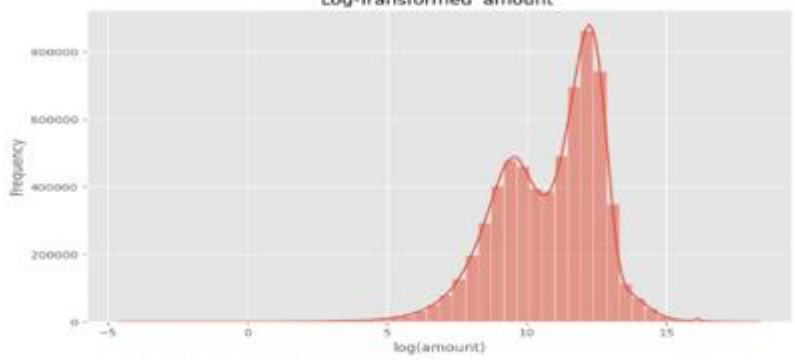


Data Preprocessing Code Screenshots

Loading Data

```
df=pd.read_csv('onlinefraud.csv')
dimensions = df.shape
print(f"Number of Rows: {dimensions[0]}")
print(f"Number of Columns: {dimensions[1]}")
```

Number of Rows: 6362620
 Number of Columns: 11

Handling Missing Data	<pre>[] df.isnull().sum()</pre> <pre> step 0 type 0 amount 0 nameOrig 0 oldbalanceOrig 0 newbalanceOrig 0 nameDest 0 oldbalanceDest 0 newbalanceDest 0 isFraud 0 dtype: int64 </pre>
Data Transformation	<pre>[] def transformationplot(feature): plt.figure(figsize=(12,5)) plt.subplot(1,2,1) sns.distplot(feature) plt.subplot(1,2,2) stats.probplot(feature,plot=plt)</pre>   <pre> minmax = MinMaxScaler() df[['amount']] = minmax.fit_transform(df[['amount']]) </pre>
Feature Engineering	Attached the codes in final submission.